Lecture 10: We talk about Bayesian linear regression now

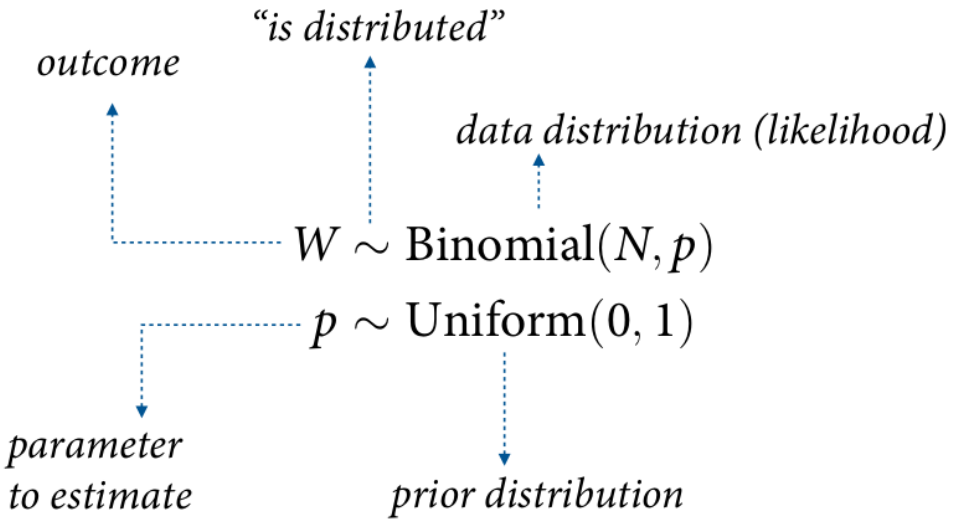
Normal data

- Remember, do not apply a procedure but build your model. Apply a strategy!
- Quite common to have normally distributed data: t-test, single regression, multivariate regression, ANOVA, ANCOVA, MANOVA, MANCOVA, etc. are all the same.
- We justified the Gaussian distribution as (i) generative (sum of fluctuations tends towards normal) and (ii) statistical to estimate mean and variance: Gaussian is the least informative distribution (or maximum entropy).
- Variables does not have to be normally distributed for a normal model to be useful.

L(earning a l)anguage for modelling

Binomial data example

- Revisit the globe tossing model:



- Just like for a language, you don't need to understand all of it. Enough to follow the flow. Add bits of understanding by revisiting the notes (SR book), the lectures, the labs.

Binomial data example

Computation can start

- Likelihood: $P(W|N,p) = \text{Binomial}(N,P)$
- Prior: $P(p) = \text{Unif}(p|0,1)$
- $\Rightarrow$ posterior: $P(p|W,N) \propto \text{Binomial}(N,P) \times \text{Unif}(0,1)$

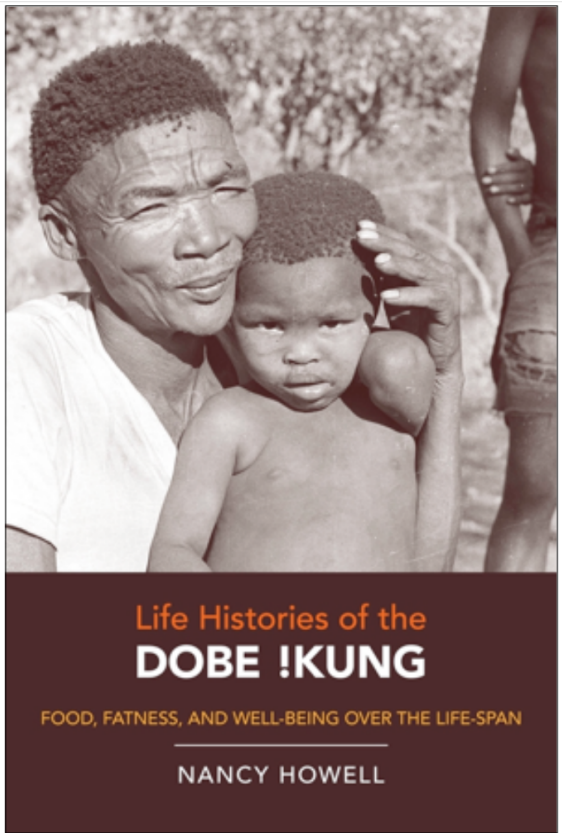
Or in R code:

```
W <- 6; N <- 9
p <- seq(from = 0, to = 1, len = 1000)
ProbaW <- dbinom(W,N,p)
Probap <- dunif(p, 0, 1)
posterior <- ProbaW*Probap; posterior/sum(posterior)
```

Linear regression

Constructing the model

- Question/goal/estimand?
- Scientific model
- Statistical model
- Validate model
- Analyse data

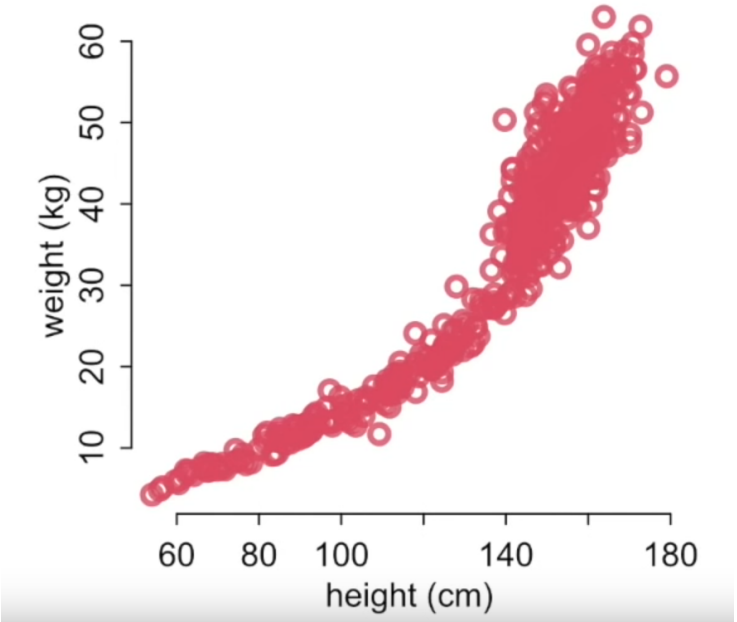


Linear regression

Constructing the model

- Question/goal/estimand?
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- Analyse data

```
library(rethinking}
data(Howell1)
```



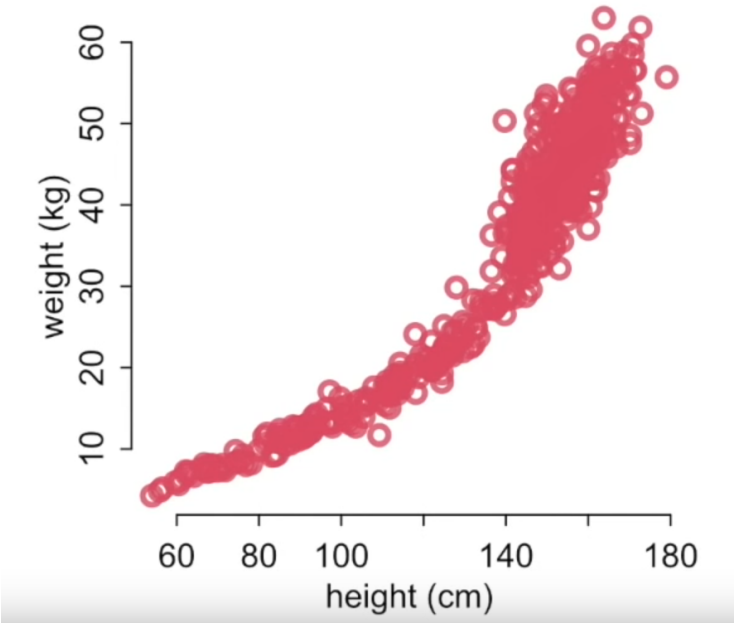
Linear regression

```
library(rethinking}
data(Howell1)
```

Constructing the model

- Question/goal/estimand?

Describe the association between weight and height.
Not simple!



Linear regression

Constructing the model

- Scientific model

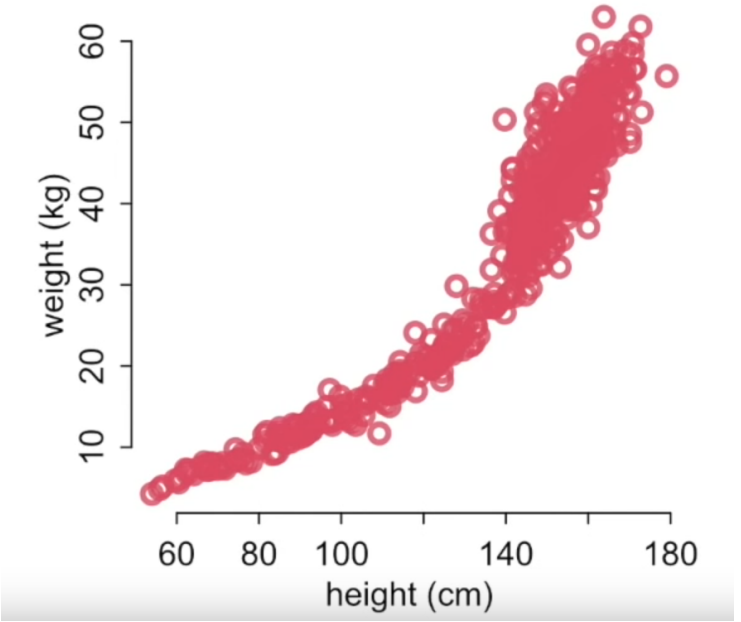
Does height influence weight?

$$H \longrightarrow W$$

$$W = f(H)$$

When we make generative models, we describe the dependency: dynamic where growth drives weight and height; Gaussian variations result from summed fluctuations.
Static point of view: height result in changes in weight; no mechanism; Gaussian variation results from growth history.

```
library(rethinking}
data(Howell1)
```



Linear regression

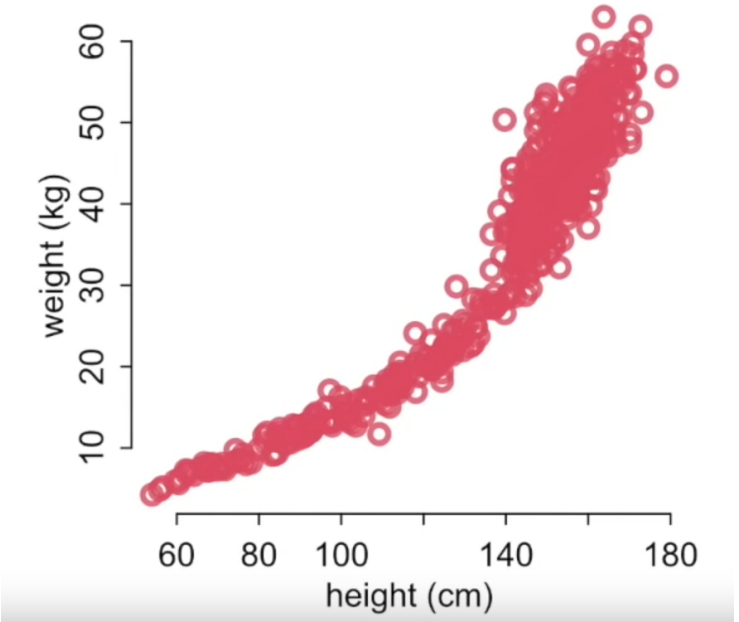
Constructing the model

- Statistical model

$$w_i = \alpha + \beta h_i + \epsilon_i,$$

with i: index, α: intercept and β:slope
In a linear regression, the equation doesn't tell you where the y values are, it tells you where the means for the y's are.
We will avoid the "ϵ_i" writing; doesn't pass to non-normal noises

```
library(rethinking}
data(Howell1)
```



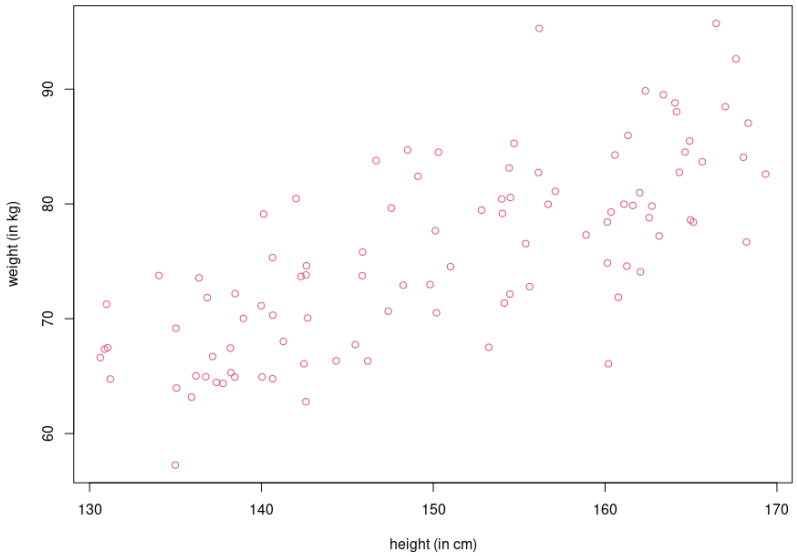
Linear regression generative model

$H \to W$

$$W_i \sim \mathrm{N}(\mu_i, \sigma)$$
$$\mu_i = \alpha + \beta H_i$$

```
alpha <- 0; beta <- 0.5
sigma <- 5
n_indiv <- 100
H <- runif(n_indiv, 130, 170)
mu <- alpha + beta*H
W <- rnorm(n_indiv, mu, sigma)
```

Synthetic individuals



Language for modelling

Normal data

- Define the variables:

$$y_i$$

$$\mu_i$$

$$\beta$$

$$\sigma$$

$$x_i$$

Language for modelling

Normal data

- Define the variables: with distributions...or mathematical equations (combining randomness/uncertainty from other variables)

$$
\begin{aligned}
y_i &\sim \text{Normal}(\mu_i, \sigma) \\
\mu_i &= \beta x_i \\
\beta &\sim \text{Normal}(0, 10) \\
\sigma &\sim \text{Exponential}(1) \\
x_i &\sim \text{Normal}(0, 1)
\end{aligned}
$$

Language for modelling

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- Note: for free, later if some x_i's are missing, we can put a distribution on those.

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Language for modelling

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$$

- Note: for free, later if some x_i's are missing, we can put a distribution on those.
- Let's start slowly and do not include any predictor yet. Intercept only regression.

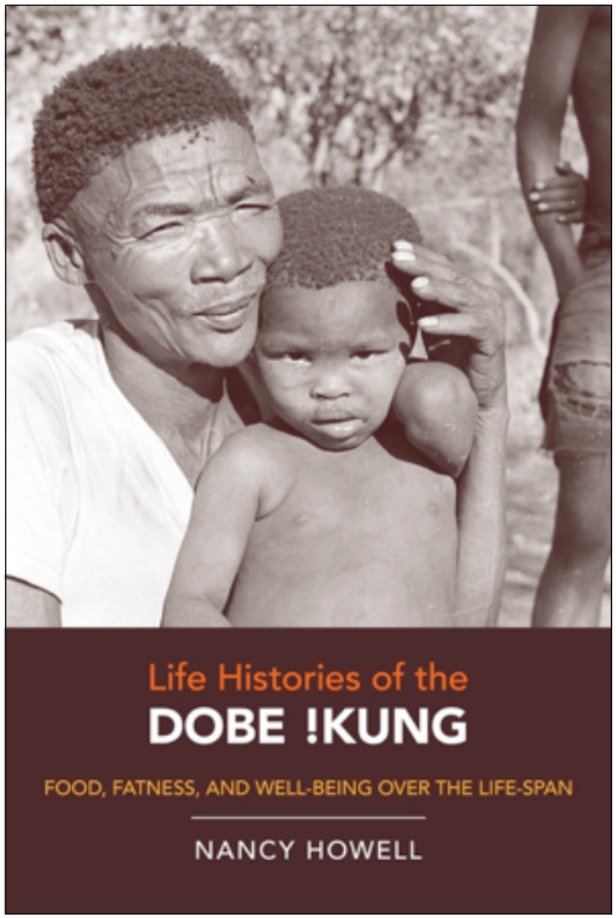
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Example data: the Kalahari foreagers data set

```
library(rethinking)
data(Howell1)
d <- Howell1
```

```
precis( d )
```

```
'data.frame': 544 obs. of 4 variables:
          mean    sd  5.5%  94.5%     histogram
height 138.26 27.60 81.11 165.74 ▁▁▁▁▁▁▁▂▁▇▅▁
weight  35.61 14.72  9.36  54.50 ▂▃▂▂▂▂▅▇▇▃▂▁
age     29.34 20.75  1.00  66.13 ▇▇▆▅▃▂▂▁▁▁▁▁
male     0.47  0.50  0.00   1.00 ▇▁▁▁▁▁▁▁▁▁▁▇
```

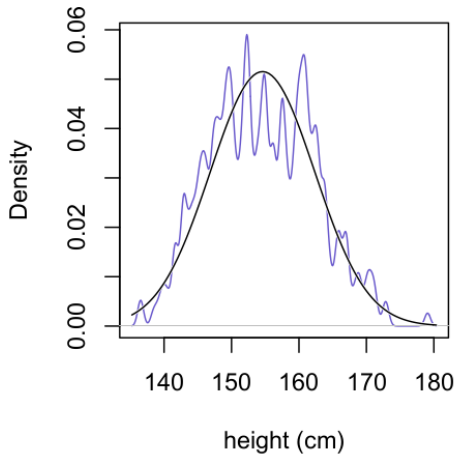


Kalahari foragers

Height Gaussian model

- First simple model:

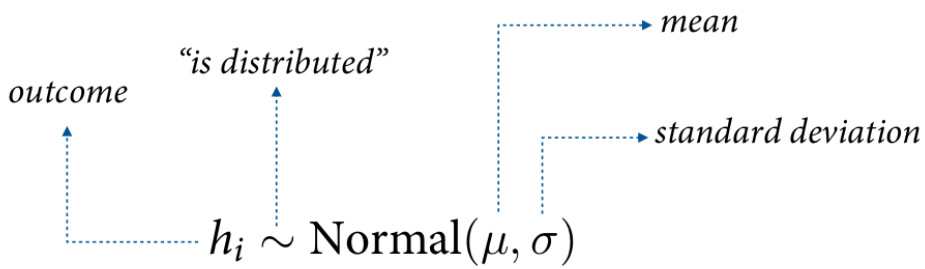
$$h_i \sim \text{Normal}(\mu, \sigma)$$



Kalahari foragers

Height Gaussian model

- Meaning of the elements of the likelihood:



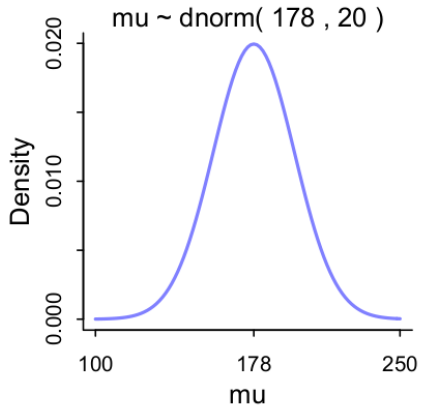
- The height h_i of each individual i is distributed normally, with mean μ (mu) and standard deviation σ (sigma).

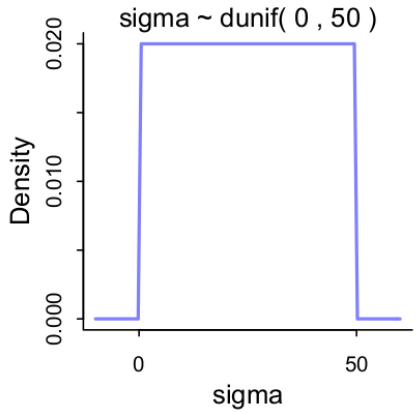
Kalahari foreagers

Height Gaussian model

- Add the priors in:

$$h_i \sim \text{Normal}(\mu, \sigma)$$
$$\mu \sim \text{Normal}(178, 20)$$
$$\sigma \sim \text{Uniform}(0, 50)$$



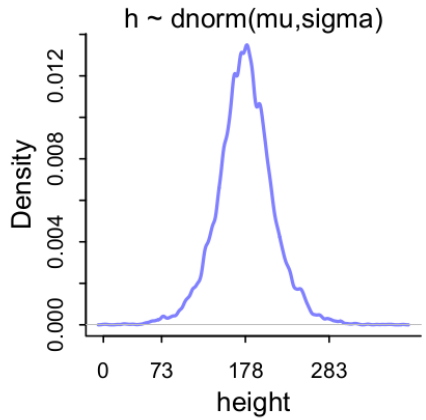


Kalahari foragers

Simulate prior predictive heights

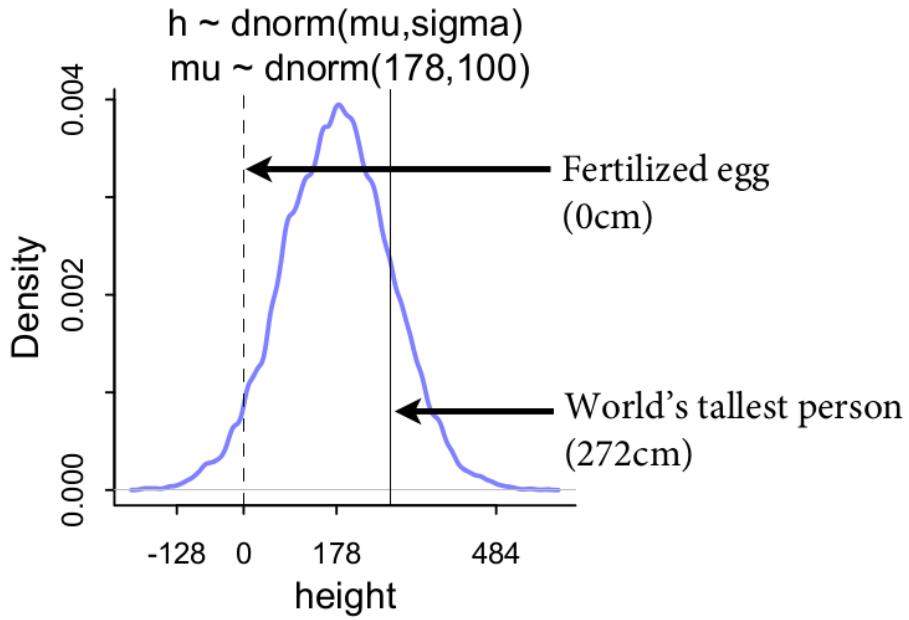
- What do these priors imply about individual heights (data) before we see the data?
- Ideas introduced last week: simulate the *prior predictive distribution*.

```
sample_mu <- rnorm(10000, mean = 178, sd = 20)
sample_sigma <- runif(10000, 0, 50)
prior_height <- rnorm(10000, sample_mu, sample_sigma)
dens(prior_height)
```



Kalahari foragers

- How sensible is my prior? Look at the implication for individuals' heights. If using $\mu \sim$ `dnorm(178, 100)` :-/ ??




```
sum(prior_height>272)
[1] 1791
sum(prior_height<0)
[1] 464
```

2D (μ, σ) Posterior approximations

1 - Grid approximation

- Remember that `d` is the `Howell1` data; `d2` is adults only.

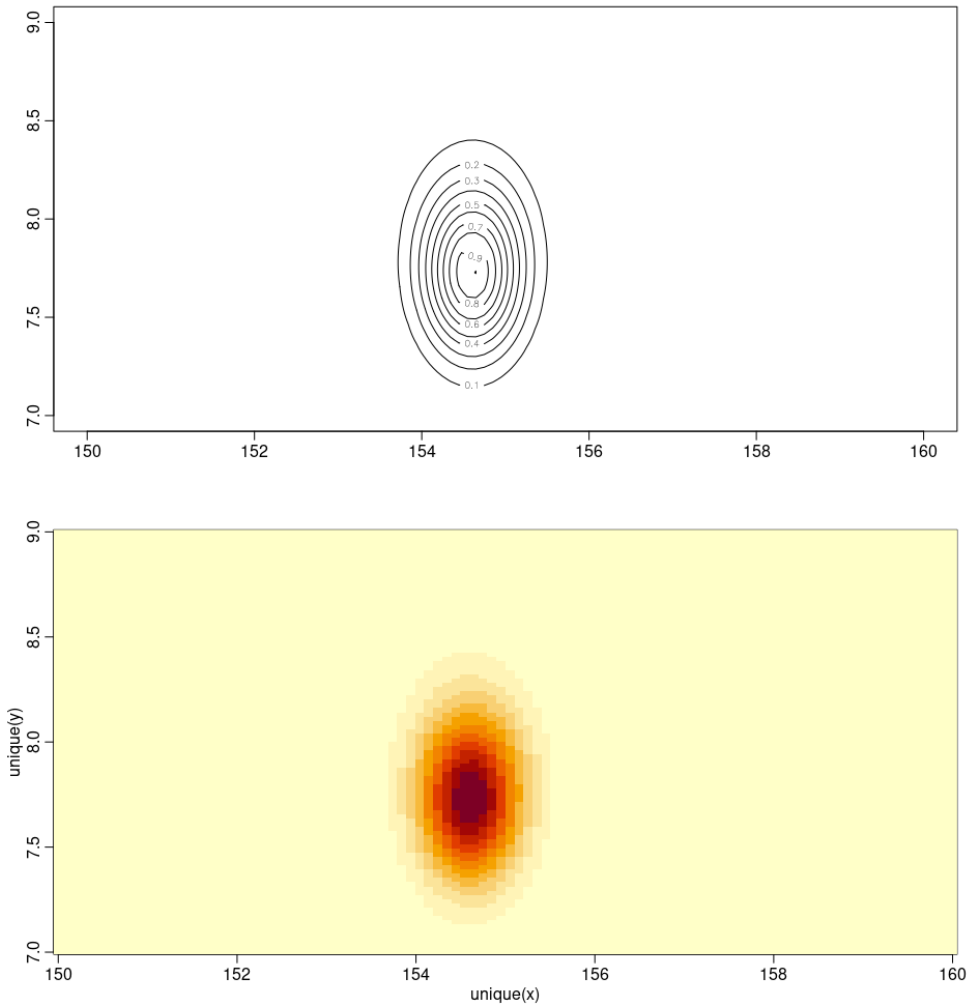
```
mu.grid <- seq(150, 160, length.out=100)
sigma.grid <- seq(7,9, length.out=100)
post <- expand.grid(mu=mu.grid, sigma=sigma.grid)
post$LL <- sapply(1:nrow(post), function(i) sum( dnorm(
    d2$height, post$mu[i], post$sigma[i], log = TRUE)))
post$prod <- post$LL + dnorm(post$mu, 178, 20, log = TRUE)
    + dunif(post$sigma, 0, 50, log = TRUE)
post$proba <- exp(post$prod - max(post$prod)) # posterior!
```

- And you can inspect this posterior via a contour plot (`contour_xyz()`) or a heatmap (`image_xyz()`)

```
contour_xyz(post$mu, post$sigma, post$proba)
image_xyz(post$mu, post$sigma, post$proba)
```

2D (μ, σ) Posterior approximations

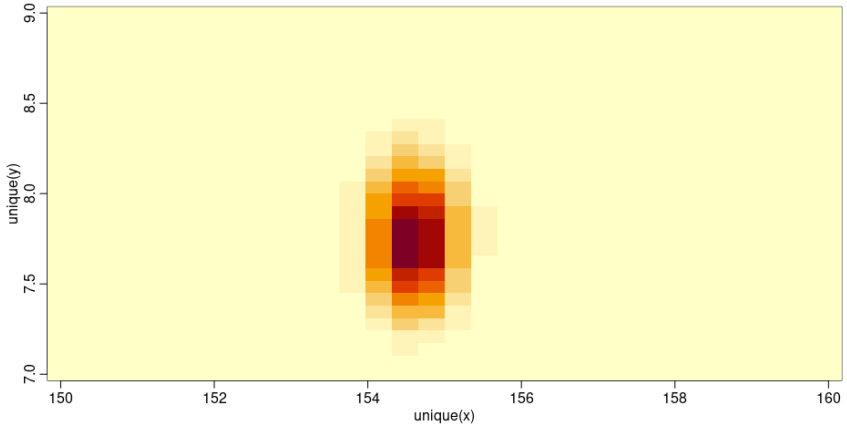
1 - Grid approximation



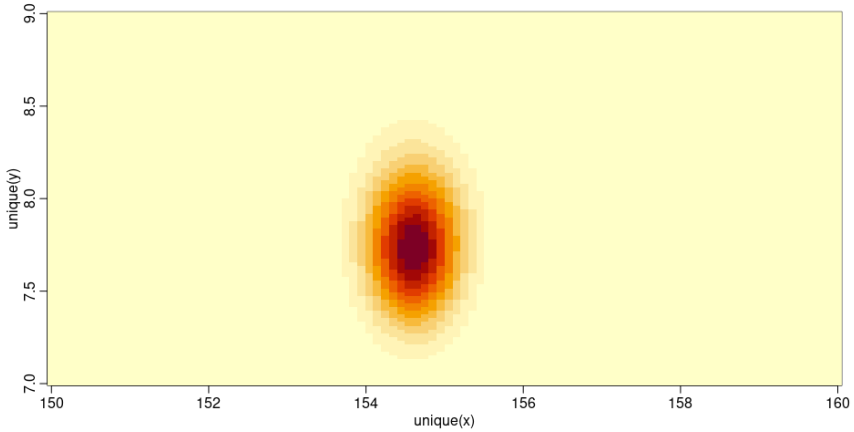
2D (μ, σ) Posterior approximations

1 - Grid approximation quality

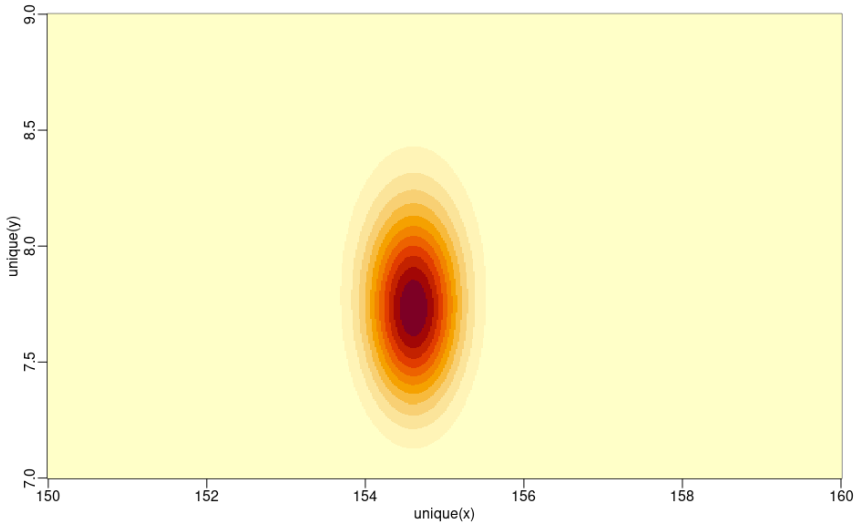
30×30



100×100



500×500



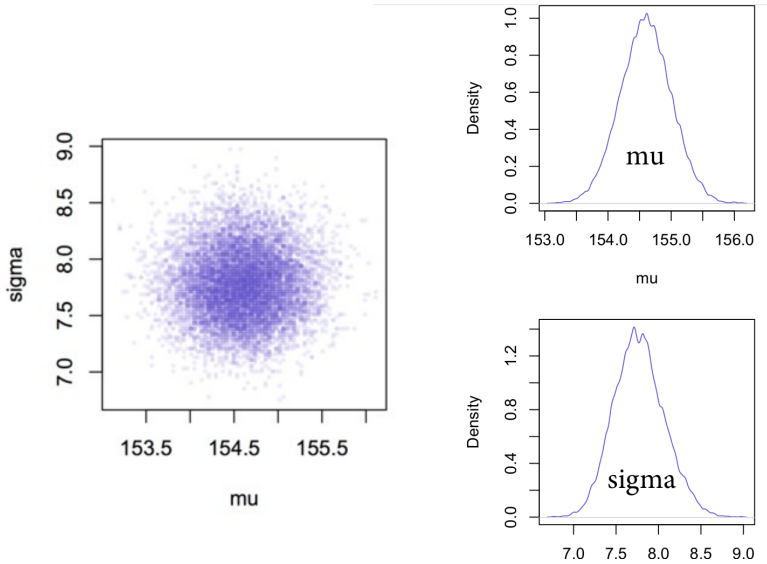
But $\sim$ 1 sec to run

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And we can sample from the posterior

Redo with different priors, different data size, different grid resolution, etc.

```
rows.sampled <- sample(1:nrow(post), size = 10000,
            replace = TRUE, prob = post$proba)
mu.sampled <- post$mu[rows.sampled,]
sigma.sampled <- post$sigma[rows.sampled,]
plot(mu.sampled, sigma.sampled, cex = 0.5, pch = 16,
            col = col.alpha(rangi2,0.1))
dens(mu.sampled);dens(sigma.sampled)
PI(mu.sampled);PI(sigma.sampled) #same-ish as HPDI
```



Next lecture

- The *quadratic approximation* with `quap()`